Test Plan: Stack Demo

Component	Details
Objectives	The goal is to ensure the core shopping flow of StackDemo works as expected: browsing and filtering products, favourites, cart, sign-in, and checkout. It should also meet performance, usability, and compatibility standards. Specific objectives include: • Verify the product listing, size filters, sorting, and product count (default 25 products) are accurate • Verify prices and totals are correct when adding, removing, or changing items in the cart • Verify sign-in, the Orders sign-in redirect, and the checkout flow work end to end • Ensure compatibility across browsers and Android/iOS devices • Verify pages load in under 3 seconds on standard connections
Approach	The test strategy combines manual and automated testing. • Manual testing will be used for usability, exploratory, and user experience testing • Automated testing will handle regression, smoke, and cross-browser tests: Selenium. • Functional, UI, compatibility, performance, and basic security testing will be conducted

Component	Details
Scope	In Scope: • Product listing, size filters, and sorting • Favourites (add and remove) • Shopping cart (add, remove, quantity, totals) • User sign-in and the Orders page • Checkout and order confirmation Out of Scope: • Native mobile apps (web version only) • Browser and OS versions older than Android 6/iOS 12 • Real payment processing and backend or admin systems
Test Deliverables	• Test cases • Defect reports • Test execution report • Final test summary report
Dependencies	• Stable, accessible test build of the site • Test user accounts and demo credentials for sign-in • Test data: product catalog (25 products), sizes, and prices • Up-to-date browsers and mobile devices (Chrome, Firefox, Safari, Edge; Android 13, iOS 16)
Test Environment	• Devices / Browsers: Chrome, Firefox, Safari, Edge (latest 2 versions); Windows 10-11, Android 11-13, iOS 14-16; desktop and mobile viewports • Server: Next.js/React • Network: Wi-Fi and mobile data (3G/4G/5G), including throttled connections

Component	Details
Risk Management	Identified Risks: • Test site or test accounts unavailable or reset unexpectedly • Site slows down or crashes under high load • Layout or behavior differences across browsers and devices Mitigation Strategies: • Keep a backup test environment and test data; use mock data if needed • Run load testing early to find performance bottlenecks • Run cross-browser tests on a device cloud and prioritize the most-used combinations
Schedule	Timeline: • Test Planning: Nov 1 - Nov 5 • Test Case Design: Nov 6 - Nov 8 • Functional Testing: Nov 9 - Nov 14 • Security and Performance Testing: Nov 14 - Nov 18 • Test Reporting: Nov 19 - Nov 26
Roles and Responsibilities	• QA Lead: Mahmoud - Overall responsibility for test execution and reporting • Test Engineers: Perform test case execution and bug reporting • Developers: Fix bugs identified during testing • DevOps: Set up and maintain the test environment